

Investing in Tech 101

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Information Technology – Overview

- Information Technology covers companies that design, manufacture and sell technology products, software and services — from chips to cloud platforms to Software-as-a-Service vendors.
- Largest sector in the S&P 500 at 39% of the index; 45+% of the MSCI Asia-ex JP Index – heavy weighting in TW and KR in Asia.
- Made up of 3 distinct subsectors, each focusing on different areas: Semiconductors, Hardware/ Equipment and Software & Services.

Focus for today



Semiconductors (19%)

- Foundational layer for all hardware Technology – every device runs on chips
- Key names: NVIDIA, TSMC, ASML, Intel, Broadcom, AMD, SK Hynix
- Broad-based demand drivers but usually indexed to the hottest product – AI servers for now.
- Cyclical industry, given heavy capital investment needs
- Geopolitically sensitive — chip supply chains sit at the centre of US-China tensions

Hardware & Equipment (10%)

- Covers physical devices, PCs, smartphones, servers, networking gear etc.
- Key names: Apple, Cisco, Dell, Hon Hai, Xiaomi
- Less capital intensive, focus on brand, channel, manufacturing efficiency.
- Products commoditise over time and margins compress
- Growth tends to be slower and tied to device upgrade cycles

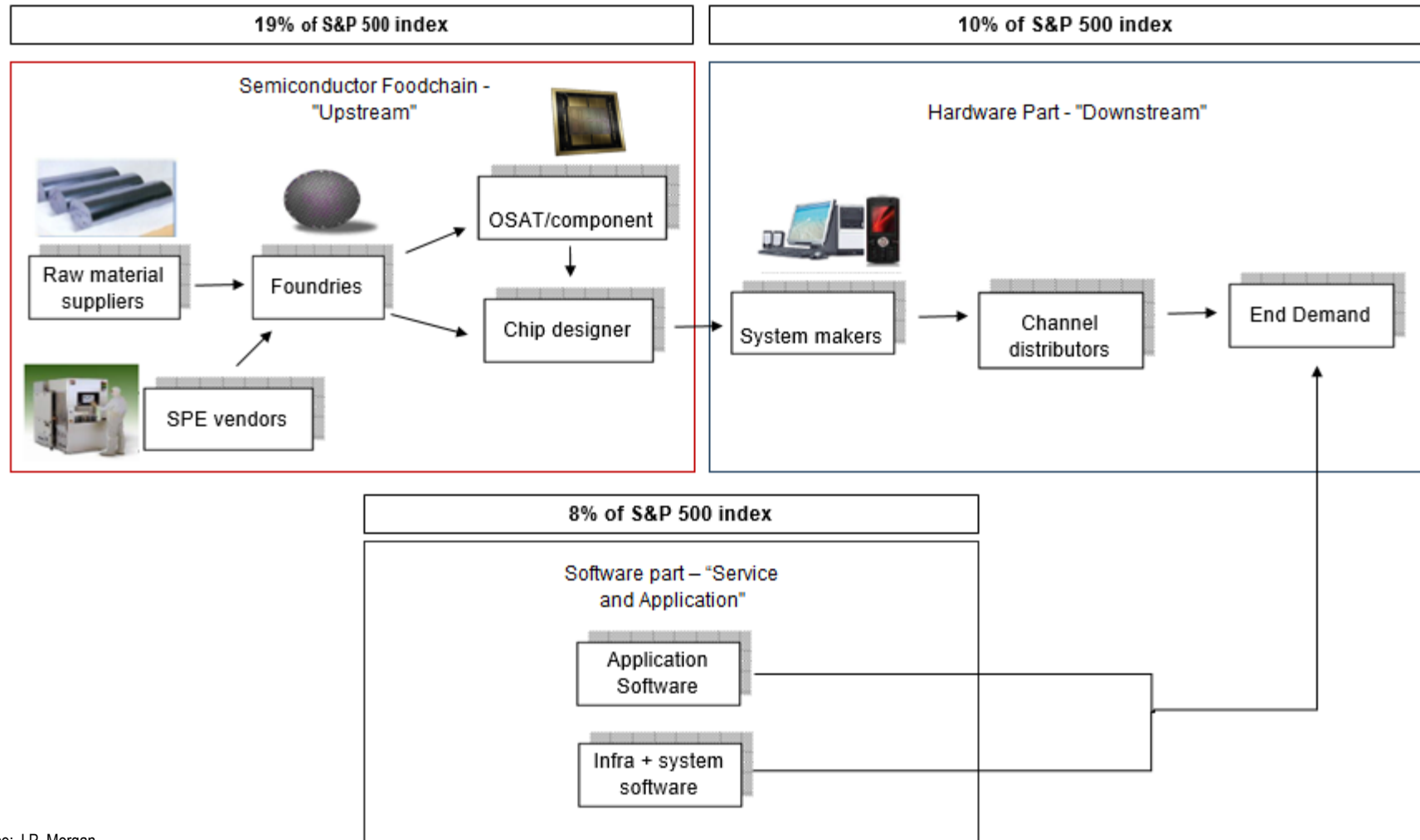
Software & Services (8%)

- Covers operating systems, enterprise applications, cloud platforms & IT services
- Key names: Microsoft, Oracle, SAP, Palantir, ServiceNow, TCS
- Asset-light model with high gross margins and recurring subscription revenue (SaaS)
- Historically valued highly by markets for earnings predictability
- Currently in the midst of GenAI disruption concerns.

Note: % is weight in S&P 500 index

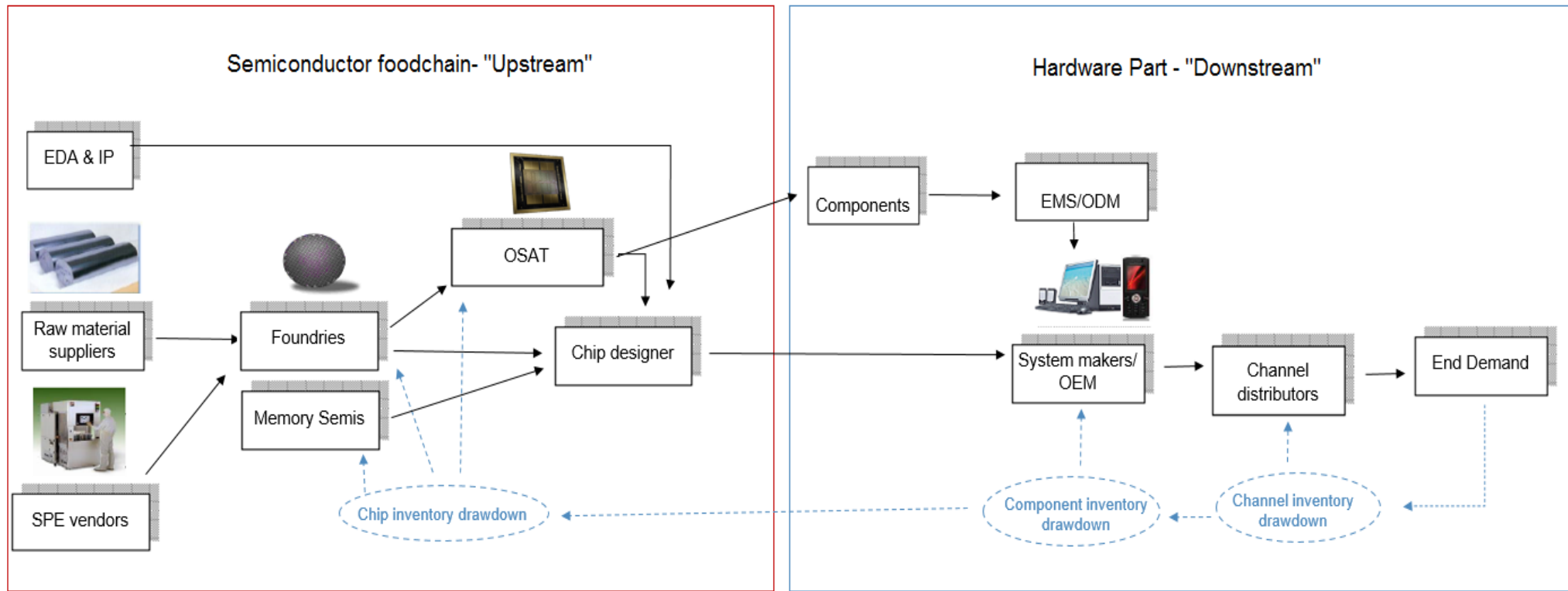
How the whole Infotech ecosystem works together

Technology Supply Chain



Source: J.P. Morgan.

Zooming In – Tech Hardware ecosystem and how business flow works



Source: J.P. Morgan.

How big is the Semiconductor market, who are the big players?

US\$14.8 trillion

Market Cap

(SOX Index, as of Jun 1st, 2026)

US\$796 billion

2025 Revenue

(Source: WSTS)

Top 10 Semiconductor Companies

Company	Ticker	Current Market Cap (US\$bn)	FY25 Revenues (US\$bn)
Nvidia	NVDA US	5,145	216
Broadcom	AVGO US	1,997	64
TSMC	2330 TT	1,894	123
Samsung	005930 KS	1,157	235
SK Hynix	000660 KS	1,082	68
Micron	MU US	1,047	37
AMD	AMD US	808	35
ASML	ASML NA	623	37
Intel	INTC US	612	53
Lam Research	LRCX US	399	18
Top 10 aggregate		14,766	886

Source: Bloomberg Finance L.P. FY26 Revenues for NVDA. Note: Shown overall corporate revenues.

Moore's Law and Semiconductors

- **Semiconductors are the building block for any tech product**
- **Moore's Law** – indicates that same semiconductor performance can be enabled at 20-25% lower cost in 2 years time OR performance can be improved 25-30% in two years, at the same cost.
- This is by far the largest productivity improvement that has been seen in any industry. The following example gives a good indicator of how much improvement has been possible.

“In 1978, a commercial flight between New York and Paris cost \$900 and took seven hours. If the principles of Moore's Law had been applied to the airline industry the way they have to the semiconductor industry since 1978, that flight would now cost about a penny and take less than one second.”

- Intel

If transistors were people



Now imagine that those 1.3 billion people could fit onstage in the original music hall. That's the scale of Moore's Law.

Transistor = "Brain Cell of a Tech Product"

More Brain cells = Greater processing power

Smaller Brain Cells = Greater processing power, in a much smaller device

Logic vs Memory semiconductors

■ Logic vs Memory semis

■ Logic chips are in charge of data computation

- CPU, SoC, GPU, ASIC, FPGA, etc. Perform computation for different types of data, algorithm, user cases, etc.

■ Memory chips handle the tasks of data storage

- DRAM, NAND Flash, NOR Flash, etc. Based on different data access speed and storage capacity

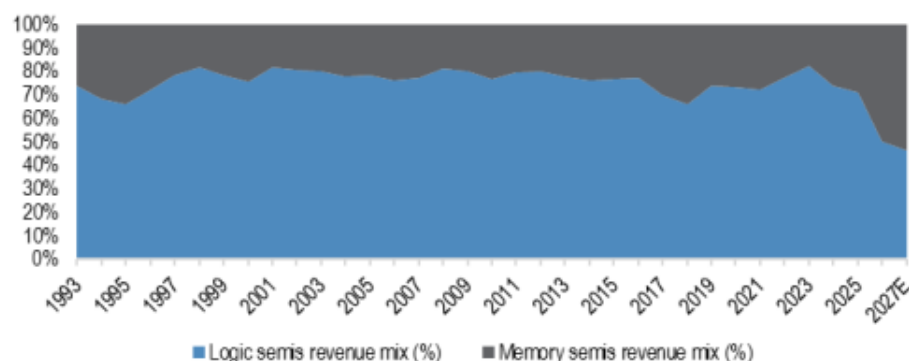
■ Global semiconductor market revenue was \$796bn in 2025 (up 26% YoY), with Memory/Logic accounting for 29%/71% of revenue

■ Memory semis cyclicalality is more volatile, due to its commodity nature

■ Logic semis market is less volatile, but still has cyclicalality, usually correlated to global GDP growth

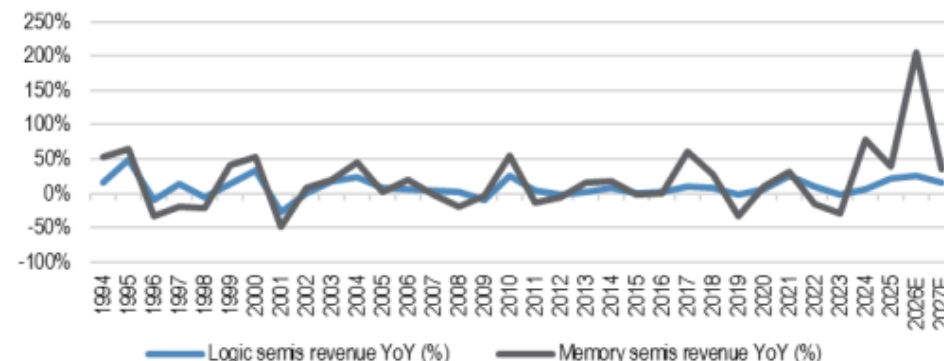
■ Fabless/Foundry business model is more prevailing in Logic semis, while key players in Memory market are IDMs

Logic vs Memory Semis Revenue Mix



Source: WSTS, J.P. Morgan estimates.

Logic vs Memory Semis Revenue YoY Change



Source: WSTS, J.P. Morgan estimates.

Semiconductor ecosystem – examples of all these kind of companies

■ Key enablers and examples of these companies

- Raw wafers: GlobalWafers, Shin-Etsu, SUMCO, Siltronic
- EDA vendors: Synopsys, Cadence, Siemens EDA
- IP vendors: Arm, eMemory, M31, Andes, Rambus
- Design services companies: Alchip, Global Unichip, Faraday
- Semiconductor Equipment vendors: ASML, LAM Research, KLA, AMAT, Tokyo Electron

■ Logic Semis

- Foundry: TSMC, UMC, GlobalFoundries, SMIC, VIS, PSMC, Intel Foundry
- Fabless Chip Designers: Qualcomm, Nvidia, AMD, Broadcom, MediaTek, Marvell
- OSAT or Assembly and Test: ASE, Amkor, JCET, TongFu, Hua Tian, KYEC.
- Power Analog: Texas Instruments, Infineon, NXP, STMicro, On Semi, Microchip

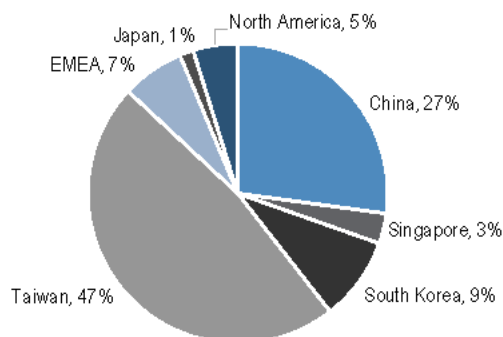
■ Memory Semis

- DRAM and NAND Flash: Samsung Electronics, SK Hynix, Micron, Nanya Tech, Kioxia

Geographic mix of the global semi ecosystem

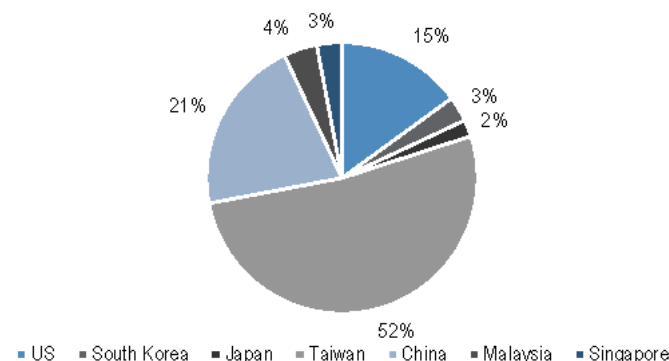
- **Foundry** – Taiwan remains the key hub with 45+% of capacity located in Taiwan
- **OSAT or Chip Assembly and Test** – Taiwan and China are key locations, together representing global 70+% of OSAT capacity
- **DRAM** – South Korea is the key market, accounting for 55% of global DRAM capacity
- **NAND** – South Korea, Japan and China together comprise 90% of global NAND capacity
- **Major IDMs** are based in Europe, US and Japan

Foundry Capacity by Region



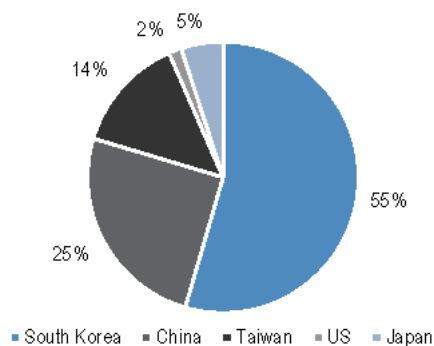
Source: Gartner® (Dec 2025).

OSAT Capacity by Region



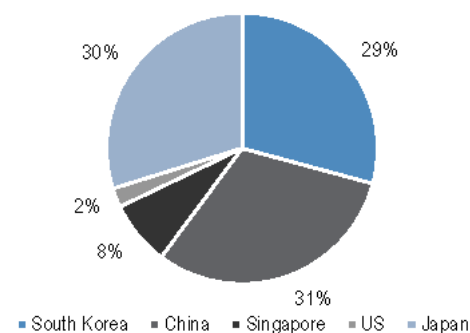
Source: CSET.

DRAM Capacity by Region



Source: OMDIA. Note: Capacity by 4Q25.

NAND Capacity by Region

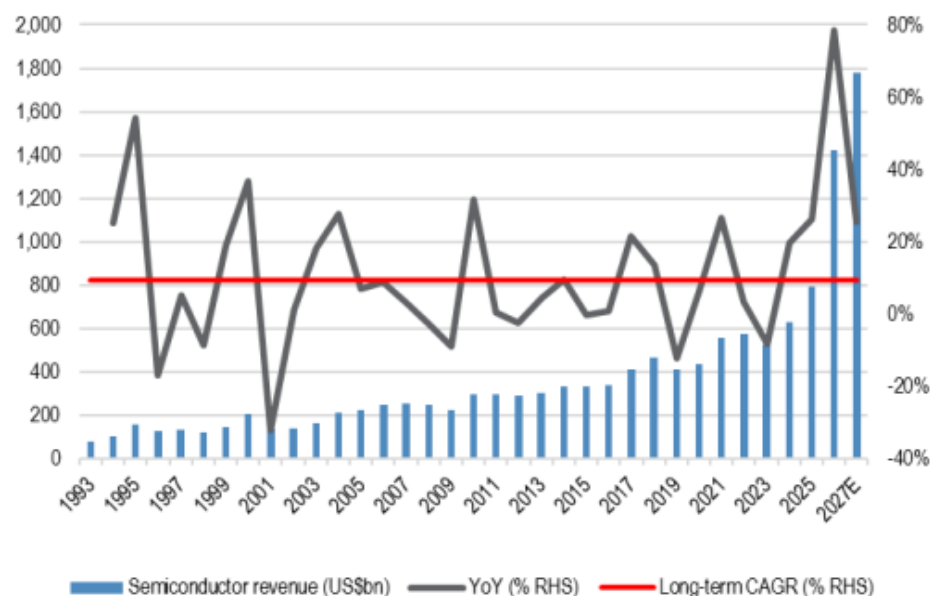


Source: OMDIA. Note: Capacity by 4Q25.

Long-term semis industry growth – Global GDP + Low Single Digits

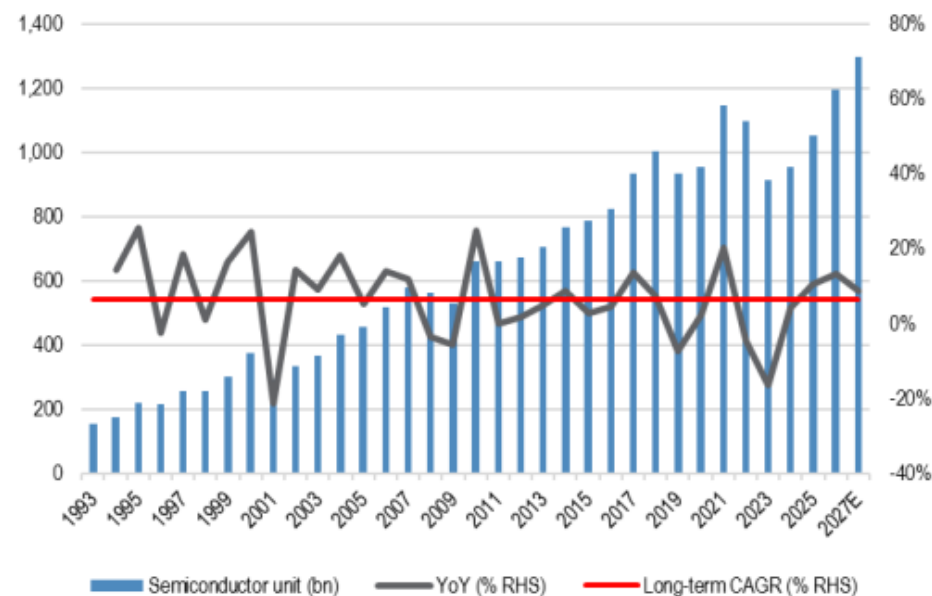
- Long-term Semis market revenue and unit growth is around 10%
- Within that we have cycles where industry growth can swing from negative to 20-30% growth, besides recent disruption from strong AI demand
- Long term, the cyclical nature is likely to remain, but amplitude of swings is getting less volatile

Overall Semis Revenue and YoY Change



Source: WSTS, J.P. Morgan estimates.

Overall Semis Unit and YoY Change



Source: WSTS.

How big is the Tech Hardware market, who are the big players?

US\$8+ trillion

Market Cap

(JPMorgan estimates, as of Jun 7th 2026)

Major Tech Hardware Companies

Company	Ticker	Current Market Cap (US\$bn)	FY25 Revenues (US\$bn)
Apple	AAPL US	4,514	416
Cisco	CSCO US	479	57
Dell Technologies	DELL US	256	114
Arista	ANET US	194	9
Seagate	STX US	190	9
Delta Electronics, Inc.	2308 TT	190	18
Western Digital	WDC US	176	10
Amphenol	APH US	171	23
Corning	GLW US	153	16
Hon Hai	2317 TT	126	261
Top 10 aggregate		6,551	715

Source: Bloomberg Finance L.P. FY26 Revenues for DELL US. Note: Shown overall corporate revenues.

Tech hardware – the key companies in the ecosystem

■ Smartphones and Consumer Electronics

- Apple , Xiaomi, Nintendo, Sony.

■ PC and Servers

- Dell, Lenovo, HPE, HPQ,

■ EMS and ODMs – Contract manufactures of all products.

- Hon Hai, Quanta, Flextronics, Celestica, Jabil, Luxshare

■ Components

- Power – Delta Electronics, Vertiv, Lite On
- Connectors – Amphenol, Molex
- Optical – Lumentum, Coherent, Innolight.

Tech Investing – Key concepts and Themes

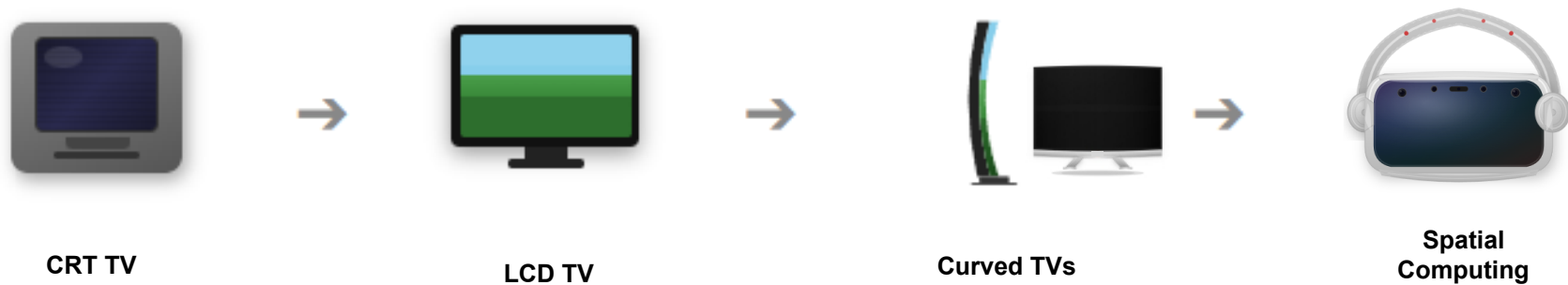
1. Tech from the lens of Digitization and Disruption

The big Tech products of today are mostly iterative digitization of human functions

Music



Digital media screen



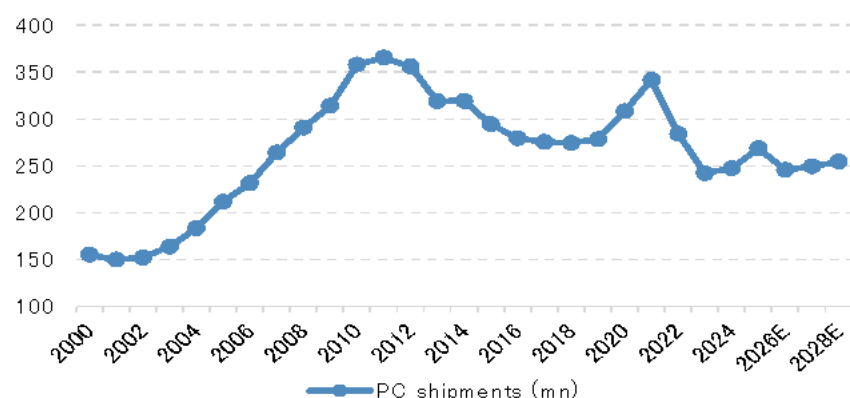
Communication and information access



2. The S-curve keeps repeating over and over for various products

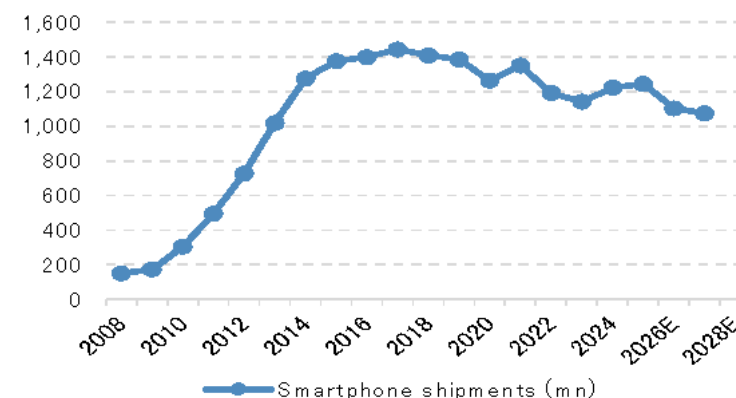
- S -Curve Framework helps to position companies and product cycles at appropriate stage of their growth
- Mispricing of stocks very likely at early and late stage of Product cycles – Market tends to think linearly.

PCs – already past the peak



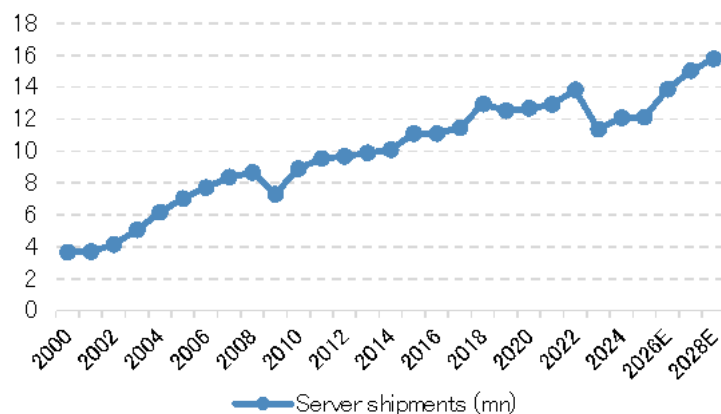
Source: Gartner®, J.P. Morgan estimates

Smartphones – already past the peak



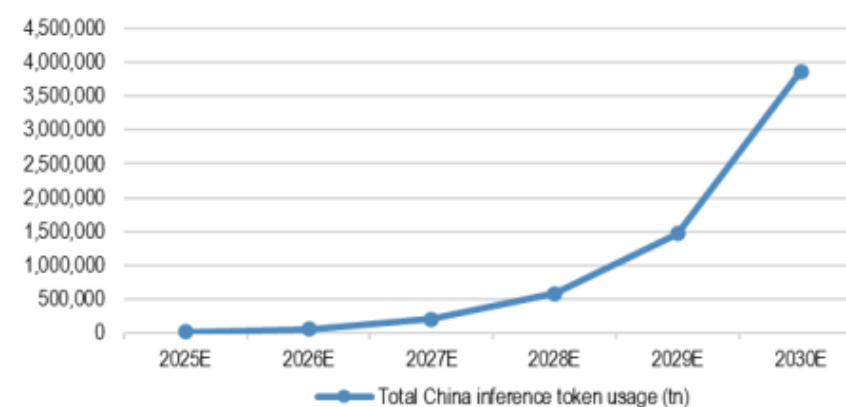
Source: IDC, J.P. Morgan estimates

Servers – In growth phase driven by AI



Source: Gartner®, J.P. Morgan estimates

AI – strong token growth driven by Gen AI adoption



Source: National Data Administration, J.P. Morgan estimates

3. There is no status-quo in Tech, much higher churn than in other industries

- Composition of Top 5 Tech companies by market cap keeps changing every 10 years.
- On average 2-3 new entrants into the top 5.

Global Top 5 in Market Cap in TMT				
Rank	2025	2015	2005	1995
1	NVIDIA	Apple	Microsoft	Microsoft
2	Apple	Google	Intel	IBM
3	Google	Microsoft	IBM	Intel
4	Microsoft	Facebook	Google	HP
5	Amazon	Oracle	Cisco	Hitachi

Asia Top 5 in Market Cap in TMT				
Rank	2025	2015	2005	1995
1	TSMC	Alibaba	Samsung	Hitachi
2	Tencent	Samsung	Canon	Fujitsu
3	Samsung	Tencent	TSMC	NEC
4	SK Hynix	TSMC	Yahoo	Canon
5	Alibaba	TCS	Hitachi	Fujifilm

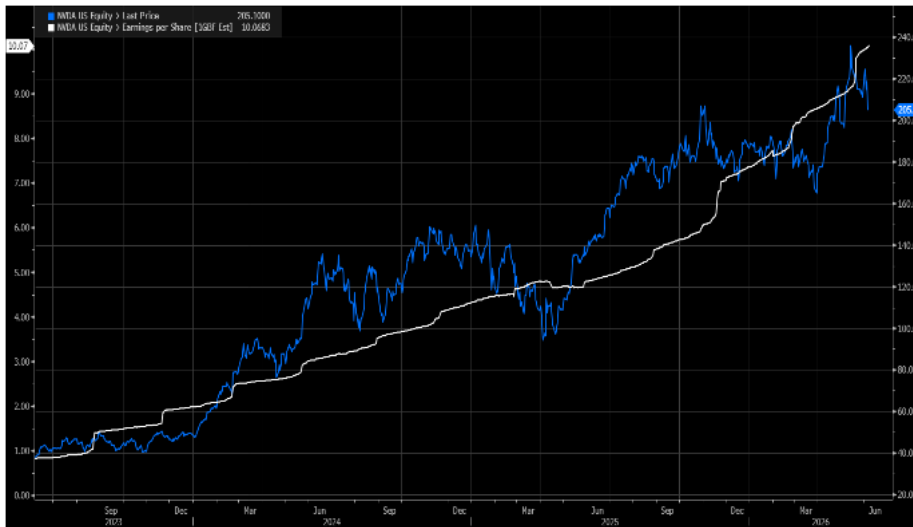
The names in red are new entrants into the top 5 list

- Much higher churn than in other industries.
- Makes comebacks very difficult due to constant change
- Gen AI is the new driver of disruption – driving value towards Semiconductors from Software vendors at this stage of the evolution.

Tech Hardware and Semis Investing – Rules of Thumb and the Three types of Tech Stocks

What drives Tech stocks? EPS revisions the most important

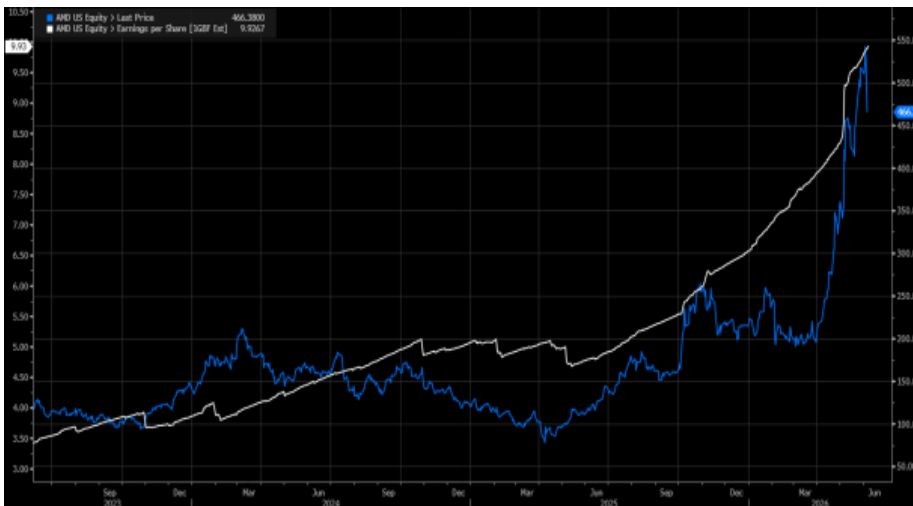
NVDA 3-year EPS ests vs share price



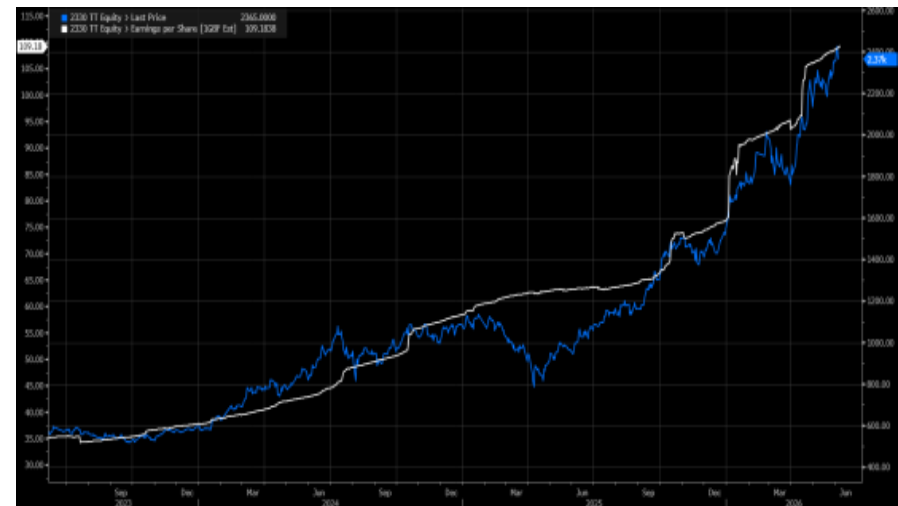
SK Hynix 3-year EPS ests vs share price



AMD 3-year EPS ests vs share price



TSMC 3-year EPS ests vs share price



Source: Bloomberg Finance L.P.

OBSOLESCENCE – the other big killer of traditional valuation concepts



Source: J.P. Morgan

- Traditional valuation concepts like trough historical price-book, trough P/E need to be taken with a grain of salt
- Book value of a DVD or printer manufacturing line may be \$500M, but the economic value may soon be zero
- Come backs in tech are incredibly difficult due to this

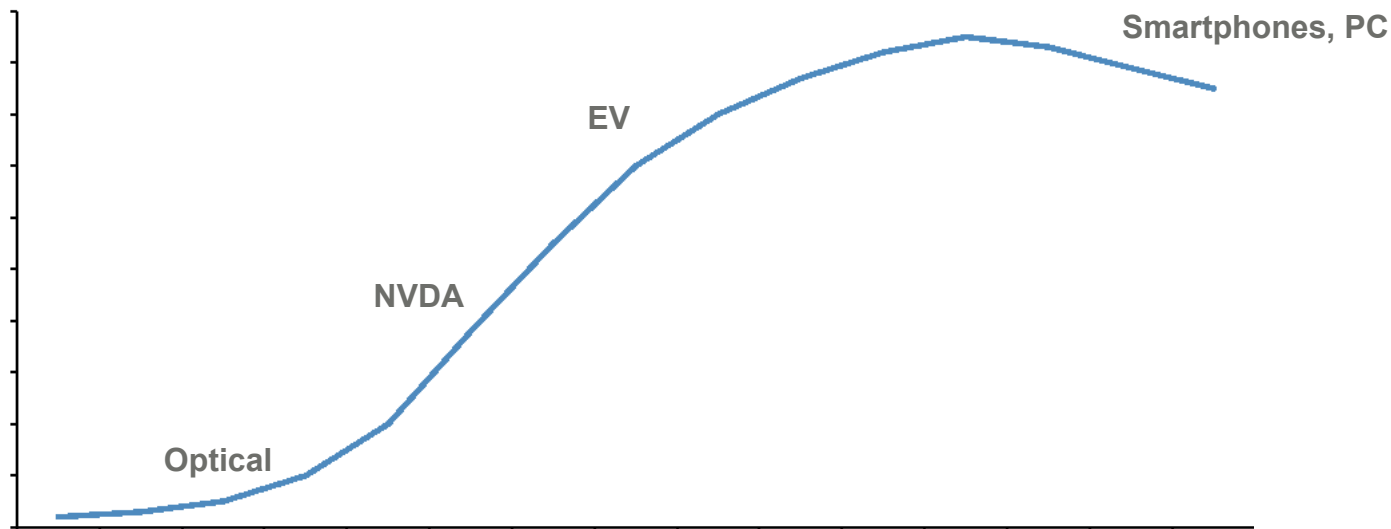
Three types of tech stocks

#1: Riding the Product cycle S-curve

#1: Product cycle stories – opportunity for multi-baggers plentiful – Boom-Bust nature usual

- Usually, component vendors or brands geared to a particular product or trend
- Important to confirm trend before entering; valuations don't really matter at all, since EPS will be always be wrong!
- Chasing the trend usually works well, trying to be contrarian usually doesn't
- Cycle doesn't turn due to expensive valuations – but some other product / trend / brand comes up to disrupt current trend

Where current product cycles fit on the S-curve



Source: JP Morgan

NVIDIA – Riding the Gen AI Adoption Curve



Source: Bloomberg Finance L.P.

QCOM (riding the 5G adoption cycle until 2022)



Source: Bloomberg

Three types of tech stocks

#2: Perfect cyclicals – supply-demand-driven stocks

#2: Perfect cyclicals or cyclical growth stocks

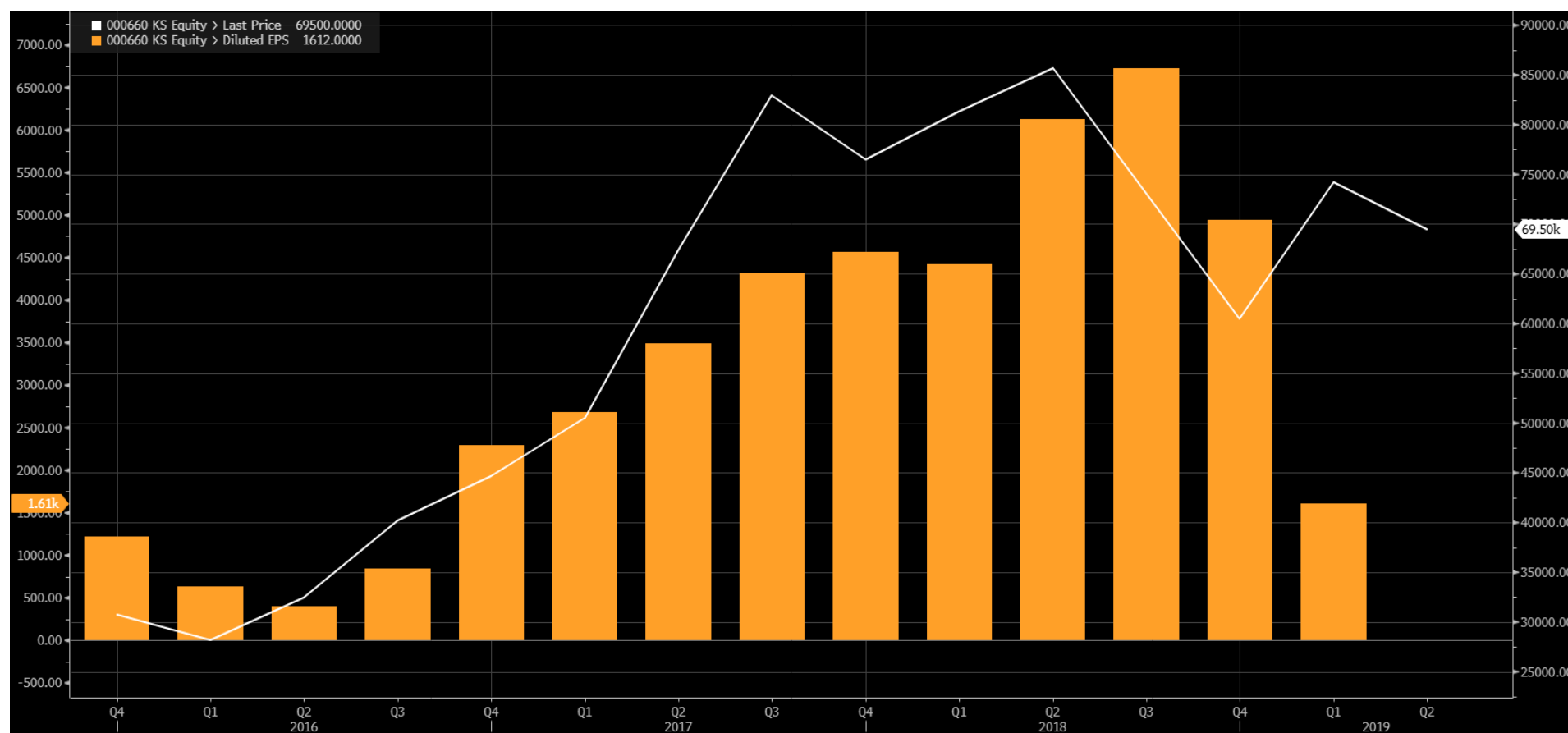
- Very similar to other cyclical sectors, such as petrochemicals or metals
- Supply-demand imbalance and time to add capacity is the key driver for share prices
- Product price (DRAM Price or Wafer Pricing) has high correlation with stocks
- Many semiconductor stocks fall in this bucket – Power Semis, Memory (DRAM and NAND Flash), Raw wafers etc.
- Buy when things are really bad – DRAM / wafers thumb rule is to buy when companies are losing money

Where are we in the cycle now?

- Memory – in a strong Upcycle – question remains – how long it will last
- Wafers – Turning around after a very long downturn
- Power Semis – Just recovering from a long 3-year downturn

Cyclical stocks – Deep drawdowns and remarkable rallies – SK Hynix (Memory)

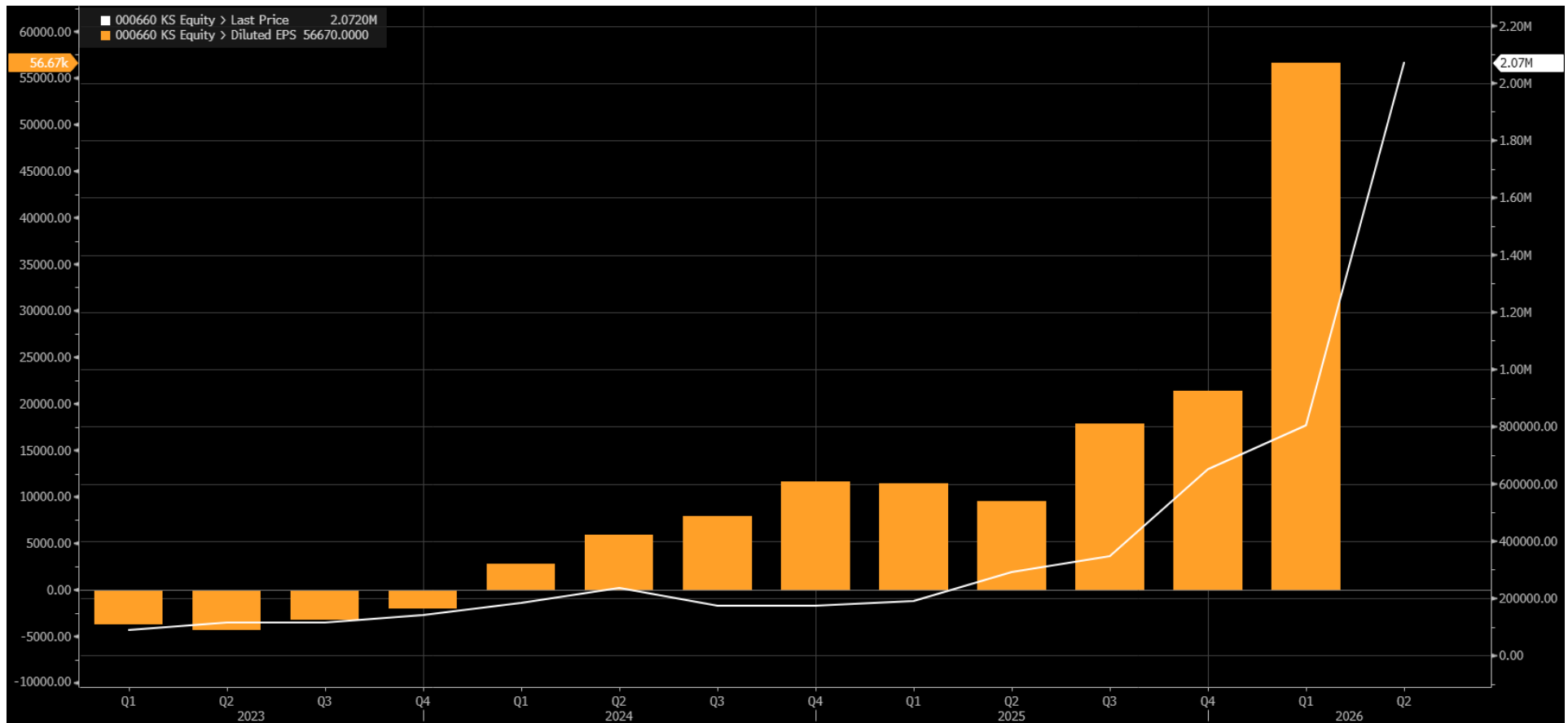
SK Hynix: share price and quarterly EPS comparison –2016-2019 – Upcycle and Downcycle



Source: Bloomberg Finance L.P.

Cyclical stocks – Deep drawdowns and remarkable rallies – SK Hynix (Memory)

SK Hynix: share price and quarterly EPS comparison – the current cycle – 2023 onwards



Source: Bloomberg Finance L.P.

Three types of tech stocks

#3: Steady growth stocks – a rare set within tech

#3: A rare commodity – steady growth stocks

- Strong free cash flows and stable margins, low capex (though not always). Strong financial management, usually steady and good dividends
- Usually own a few key technologies; not too dependent on any one product or end market – TSMC makes 95% of AI chips, but their AI exposure is still only <30%.
- Usually, the shovel sellers for the gold diggers in any new product cycle – so can dodge the extreme volatility of product cycles
- Examples – Delta, TSMC, ASML

Who are these candidates in Asian Tech?

- **Delta** – Power supply and solutions vendor for all verticals / end markets
- **TSMC** – Semiconductor manufacturer / technology provider for anyone without a silicon fab
- **ASML** – Leading semiconductor equipment vendor with a monopoly position in advanced lithography

How do we value Tech Hardware and Semiconductor companies?

■ Primary Valuation Tool — 12 Month Forward P/E

- Most widely used metric across the sector.
- Forward P/E cuts through near-term cycle noise better than trailing
- Semis are cyclical — earnings swing heavily peak to trough, making trailing P/E misleading

■ P/Book for extreme cyclicals, especially at cycle turns

- At cycle turns, P/E won't help, as cos. start to over/under earn
- P/B becomes anchor, esp. at the bottom of a cycle.

■ DCF – Reverse DCF more helpful than straight DCF

- Volatility in earnings and cashflow makes it less useful.
- Reverse DCF to verify what's priced in – a handy tool.

■ But – EPS revisions >>> Valuation !

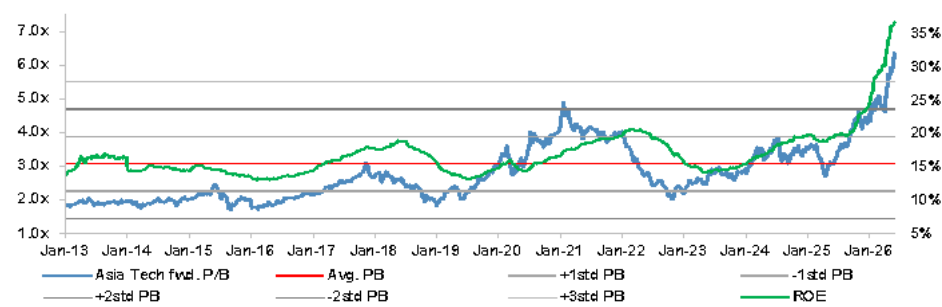
- EPS revision momentum trumps valuations
- Inventory levels, Orders vs Shipments also help

Asia Tech 12M fwd. PE valuation



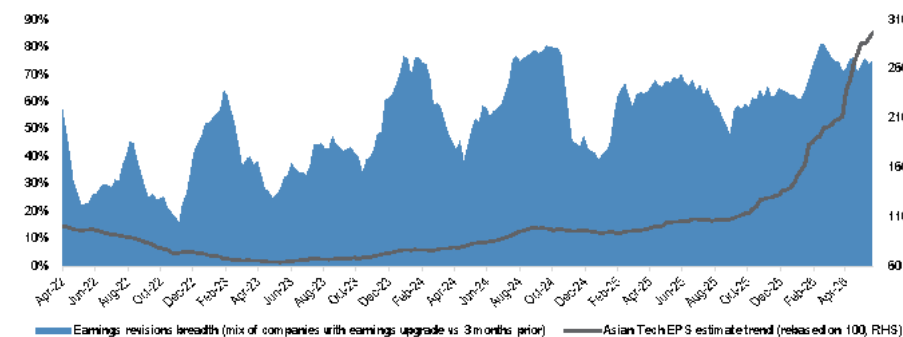
Source: Bloomberg Finance L.P., J.P. Morgan

Asia Tech 12M fwd P/Bk valuation with ROE



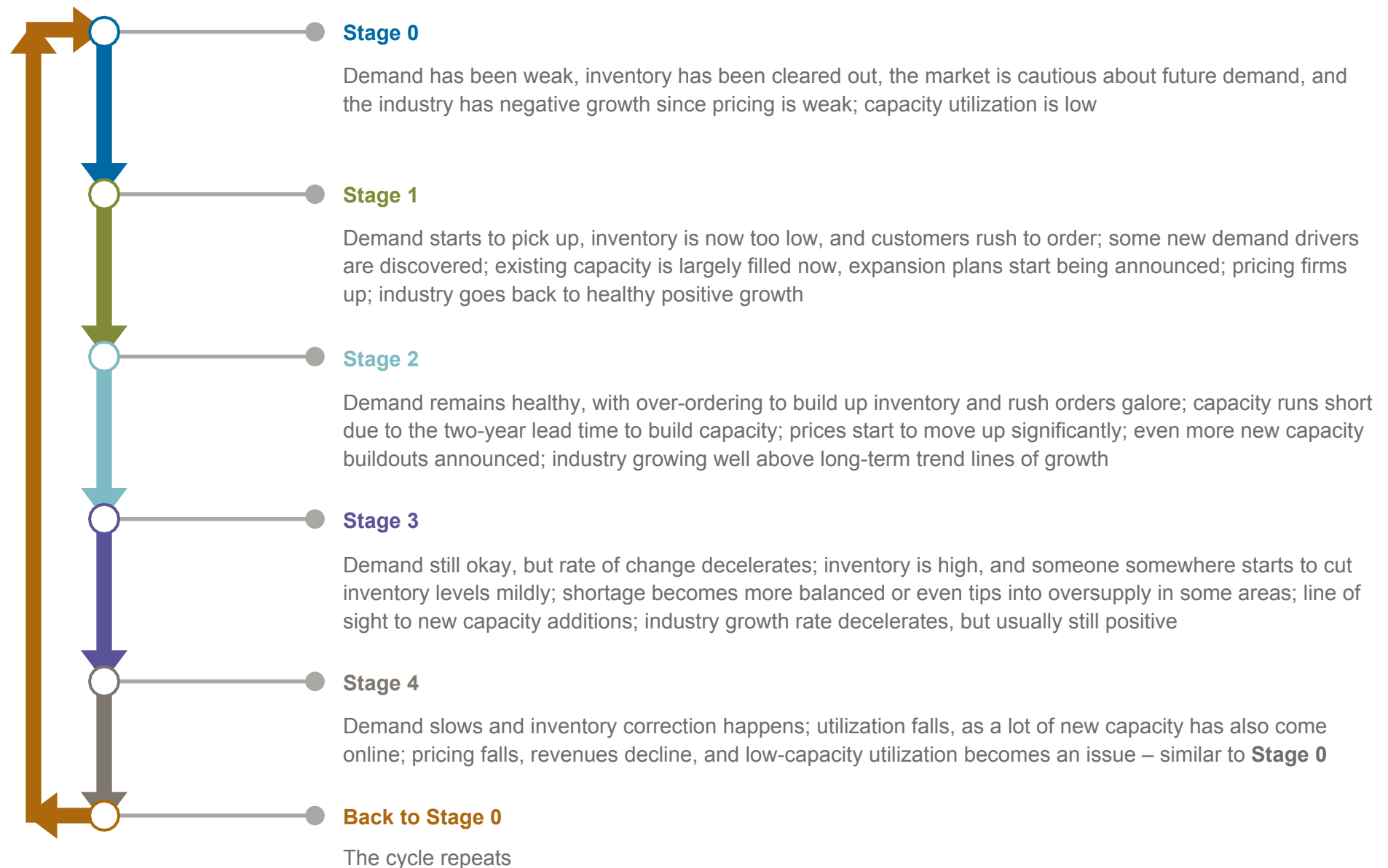
Source: Bloomberg Finance L.P., J.P. Morgan.

Asia Tech EPS Revisions



Appendix

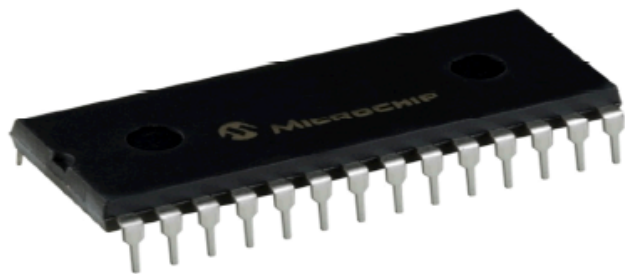
Anatomy of a typical semi cycle



What is a Semiconductor?

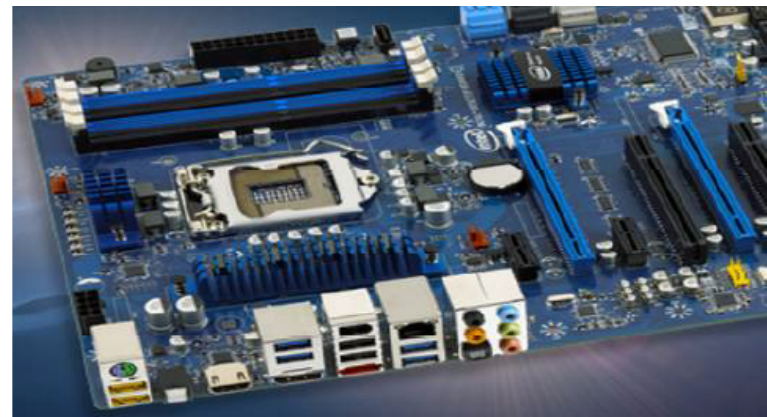
- There are 3 categories in electricity field based on resistance they provide for the passage of current through them: (1) conductors (metals such as gold, copper etc.), (2) insulators (plastics, rubber etc.), and (3) semiconductors.
 - Semiconductors, as the name suggests, have their conductivity in between that of conductors and insulators.
- Why are semiconductors so important?
 - Engineers try to design an apt blend of conductors, semiconductors, and insulators to precisely control the flow of current through the circuit.
 - The required functionality can be derived from an electronic circuit, if the passage of current can be controlled through it.
 - Hence, the "semi-conductive" nature of semiconductors makes them apt for any circuit design.
- Transistors are the building blocks of an IC (Integrated Circuit)
 - As DNA is the basic building block of genes in a human body, transistors are the building blocks of any semiconductor circuit.
 - Transistors can act as a “power switch”, which allows the current to flow through it when it is in "on" state, while no current can pass through it when it's in "off" state, illustrating their ability to switch roles between conductors and insulators.

A Packaged Integrated Circuit



Source: Microchip.

A PC Motherboard

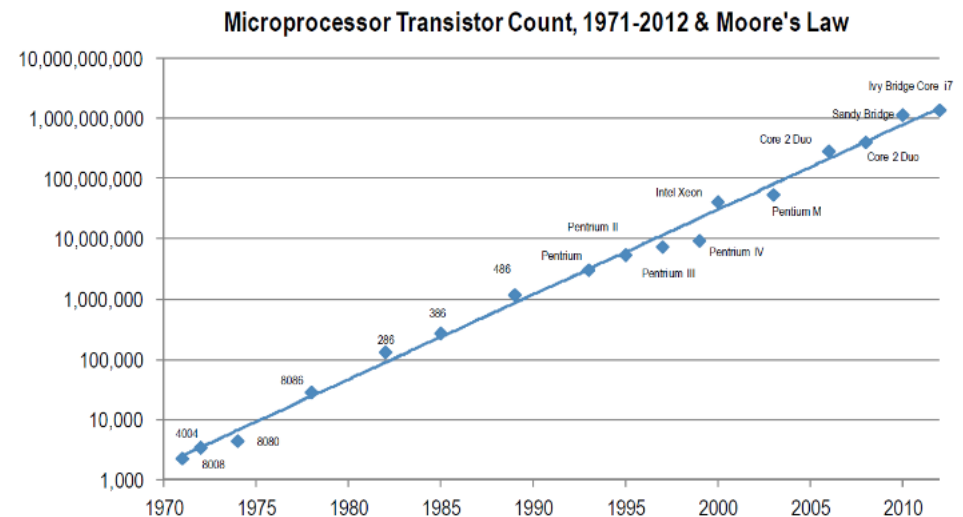


Source: Intel.

What is Moore's Law?

- Moore's Law – Gordon Moore, founder of Intel, put forward an interesting observation that the number of transistors on a chip usually doubles every two years.
- Semiconductor lithography is used to print transistors on a silicon wafer, termed as etching in semiconductor industry.
 - Foundries coat the wafer with a photoresist material and expose it to an optical light beam, hence etching the required circuitry on the wafer.
 - This process is repeated over and over in a wafer fabrication process (depending on how many layers of an IC) and is the most important part of building integrated circuits on a wafer.
- **What does process migration (such as moving from 7nm to 5nm) actually mean?**
 - To increase die output per wafer, we can print more transistors per wafer.
 - Now, there are two possible methods to do this:
 - 1) Foundries increase the wafer size, thus providing more area for printing, but this is not very viable as Fabs and lithography tools are designed for a particular wafer size. Wafer size migrations typically happen once in 12-15 years, if not longer.
 - 2) Technology advancements (process node migration) be made to reduce the size of transistor to fit in more transistors on same wafer area.
 - **The second option is usually adopted in the semiconductor industry and essentially brings Moore's Law into discussion.**

Moore's Law

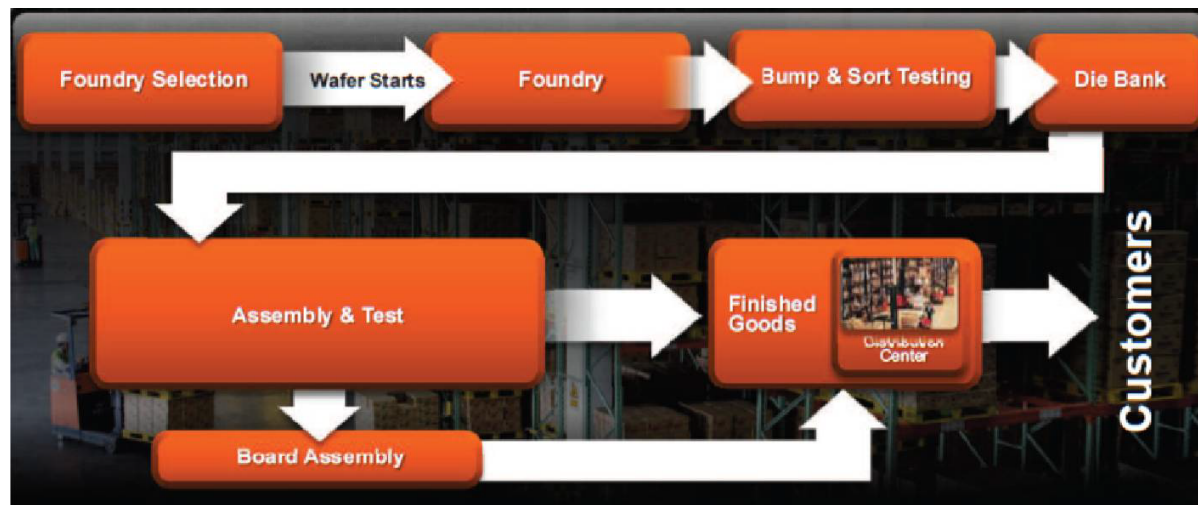


Source: Intel Corp.

IC Fabrication – the Pizza Analogy

- We can draw a close analogy of pizza making process to semiconductor manufacturing.
 - If we were to look at the pizza making process, the chef usually starts with a circular pizza base, tops it up with the desired topping, bakes it and finally cuts it into required number of slices.
 - Once the circuit design is firmed up (pizza flavor being chosen), transistors are typically printed on a silicon wafer in the desired pattern. To do this, raw wafers (pizza dough) are taken through a step of processes called photolithography and deposition which results in materials being deposited and circuits being built on the raw wafer (pizza being baked). The lead time for which is usually 2+ months.
 - These silicon wafers are then sliced into small rectangular pieces, generally known as dies, dependent on how many transistors should be present on every die. Dies are then mounted on a plastic or ceramic base.

Semiconductor Supply Chain Block Diagram

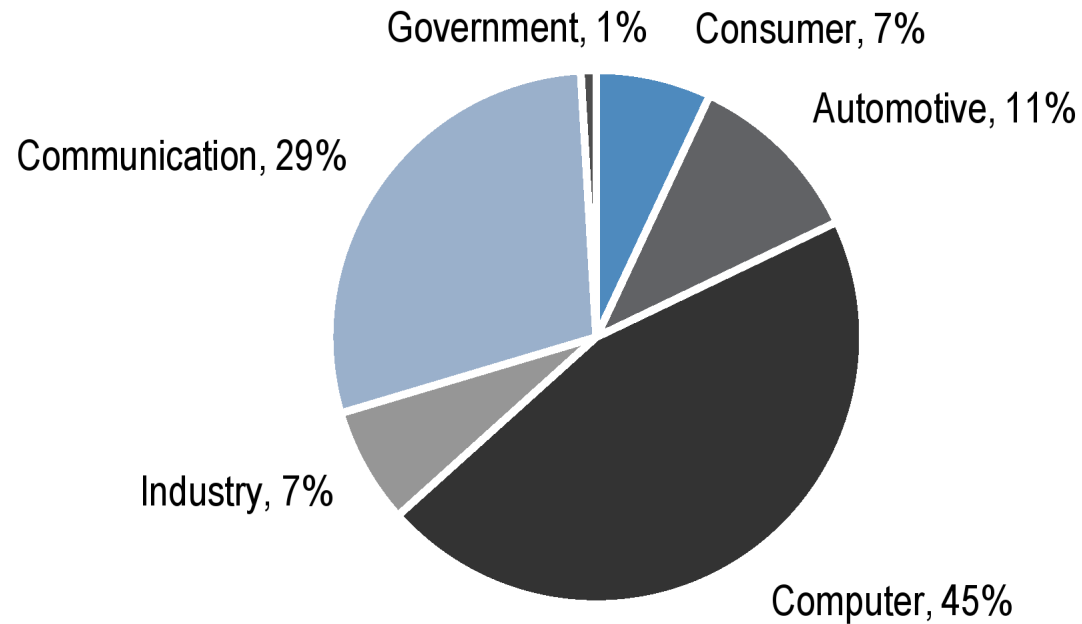


Source: AMD.

Key end markets for semis – they are all pervasive

- PC, smartphone and AI servers are key demand markets for semis
- Automotive remains a small portion within total semis market, with only 11% contribution

2025 Semis Revenue by End Application



Source: WSTS.

Geopolitics and semis – An increasingly in-focus topic

■ Can the Global Supply Chain be re-shored to US and Europe?

- Efforts are on, but short answer is No – very difficult and expensive
- Reshored fabs face significantly higher operating costs than Asian counterparts

■ China Localization and indigenous semi ecosystem

- More progress than expected – domestic self-sufficiency moving fast
- Mature node technologies increasingly being localized
- Leading edge chips significantly behind

■ Semis is a globalized industry where each participant brings specialization at a global level

- ~80% of Chip design revenue is in the US
- ~90% of Chip Design Tools Software is from the US
- Some of the most intricate Semiconductor equipment and material comes from Netherlands and Japan
- Korea dominates in memory
- Taiwan dominates in Logic Foundry
- China still dominates as an intermediate consumer of chips (due to a large percentage of electronic assembly)
- Cost of breaking this efficient structure is very high – and incremental investment may not earn any ROIC

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